Lab 2 – GradMap SRS Document

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Sections 1 & 2

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1. Introduction

Selecting and arranging college courses each semester is a critical task for students, that it remains one of the most time-consuming and error-prone aspects of academic planning. Students must interpret complex degree requirements, keep track of prerequisite chains, fit classes into limited time slots, and balance academic responsibilities. Advisors, meanwhile, manage large caseloads of students often lack sufficient tools to efficiently review, correct, and approve schedules. These challenges can lead to delayed graduations, unnecessary stress, and inefficient use of advising resources. These are all issues that collectively impact student success at a grand scale.

1.1 Purpose

GradMap is a web-based academic planning application designed for university students, academic advisors, and system administrators. The system provides separate interfaces for each user role to support academic planning, schedule management, and advisor review. The student interface is optimized for standard web browsers and allows students to create and manage their academic schedules, while the advisor and administrator interfaces support schedule review, academic guidance, and system management.

Key features for students include the Degree Progress Tracker, where students can view the requirements for their selected degree program and monitor their progress toward graduation. The Schedule Planner allows students to organize courses by semester and generate academic plans based on program requirements. Students can also view course details and adjust their schedules as needed to maintain progress toward completing their degree requirements.

GradMap's key features for advisors include the Schedule Review dashboard, where advisors can access student-generated schedules, review academic plans, and provide feedback or approval. Advisors can leave notes or suggestions to guide students in adjusting their schedules to remain on track for graduation. Administrators have access to system management features such as the Course Catalog dashboard and the Degree Requirements manager. These tools allow administrators to maintain course information, update program requirements, and manage user accounts within the system.

GradMap is not a university course registration system. Students cannot register for courses directly through the application and must complete official registration through the university's existing enrollment systems. GradMap is also not intended to replace academic advising; rather, it functions as a planning tool that helps students organize their academic paths and enables advisors to review and guide those plans more efficiently.

1.2 Scope

GradMap is a web-based academic planning system designed to assist university students in planning and managing their degree progress. The system enables students to generate course schedules based on degree requirements and track their progress toward graduation. In addition to supporting students, the system allows academic advisors to review student schedules, provide feedback, and approve academic plans. GradMap allows students to create and manage personalized degree plans while displaying course requirements for specific academic programs. Advisors can access student-generated schedules to review and approve them, ensuring that students remain on track toward completing their degree requirements. The system also

maintains a centralized database containing course information and degree requirement data to support accurate academic planning. GradMap is not intended to replace existing university course registration systems. Instead, it serves as a planning and advising tool that helps students visualize and organize their academic progress before completing official course registration through the university's existing enrollment systems.

1.3 Definitions, Acronyms, and Abbreviations

2. **Academic institution**: An organization that provides higher education programs and degrees
3. **Academic pathways**: Structured sequences of courses leading toward degree completion
4. **Academic planning**: The process of organizing coursework across semesters to satisfy degree requirements
5. **Apache**: an open-source web server used to deliver web content
6. **Authentic workflows**: Real-world user processes performed in operational academic settings
7. **Continuous Deployment (CD)**: The automated release of software updates to a production environment
8. **Continuous Integration (CI)**: The automated testing and integrated of code changes.
9. **Cross-platform**: The ability of software to operate on multiple operating systems such as Windows, macOS, and Linux
10. **Database**: A structured collection of data stored electronically
11. **Git**: A distributed version control system for tracking code changes
12. **GitHub**: A web-based platform for hosting Git repositories and supporting collaboration

13. **GitHub Actions**: Automation tools within GitHub used for CI/CD processes
14. **GitHub Workflows**: Configured automation pipelines defining CI/CD tasks
15. **Integrated Development Environment (IDE)**: Software that provides tools for coding, testing, and debugging applications
16. **Java**: An object-oriented programming language
17. **MySQL**: An open-source relational database management system
18. **Nginx**: A high-performance web server and reverse proxy
19. **Querying**: The process of retrieving specific information from a database
20. **Server-side Language**: A programming language that executes on a server rather than a client device
21. **Version control**: The practice of tracking and managing changes to source code
22. **Visual Studio Code (VS Code)**: A code editor used for software development
23. **Web-based application**: Software accessed through a web browser rather than installed locally

1.4 References

1. Coursedog. "Academic Scheduling Software." *Coursedog*, www.coursedog.com/ppc/d-scheduling-1cta?utm_source=bing&utm_medium=cpc&utm_campaign=NA%2BNonBranded%2BBing&utm_agid=1349103273937169&utm_term=university+scheduling+software&matchtype=p&creative=&device=c&placement=&msclkid=150f2b848aa2174d9b70435e5422ac86. Accessed 29 Sept. 2025.

2. "Common Issues with Degree Works Audits (and How to Fix Them)." *David Kent Consulting*, 26 Sept. 2025, davidkentconsulting.com/blog/common-issues-with-degree-works-audits-explained/. Accessed 06 Oct. 2025.

3. CognitoForms. *CognitoForms*, https://static.cognitoforms.com/content/support/assets/images/basicadmissionsapplicationworkflow-0f52033c-6cfe-42af-84b7-f27227f9aa36-dc49634-1500x650.png. Accessed 29 Sept. 2025.
4. South Texas College. "DegreeWorks FAQ." *South Texas College*,

studentservices.southtexascollege.edu/advising/degreeworks/faq.html. Accessed 28 Sept. 2025.

5. Searcy, Lucy. "Embracing Technology in Academic Advising: A Vantage Point from 14 Years in Higher Education." *NACADA The Global Community for Academic Advising*, 15 June 2025, nacada.ksu.edu/Resources/Academic-Advising-Today/View-Articles/Embracing-Technology-in-Academic-Advising-A-Vantage-Point-from-14-Years-in-Higher-Education.aspx.

1.5 Overview

The remainder of this Software Requirements Specification (SRS) document provides a structured description of the GradMap system and its requirements. Section 2 presents the overall description of the system, including the product perspective, major system functions, user roles, and any constraints or dependencies that may affect development. These sections provide developers with a high-level understanding of how the system operates and how different components interact. Later sections of the document will provide more detailed specifications of the system's functional requirements and expected behaviors to guide the implementation and testing of GradMap.

2. Overall Description

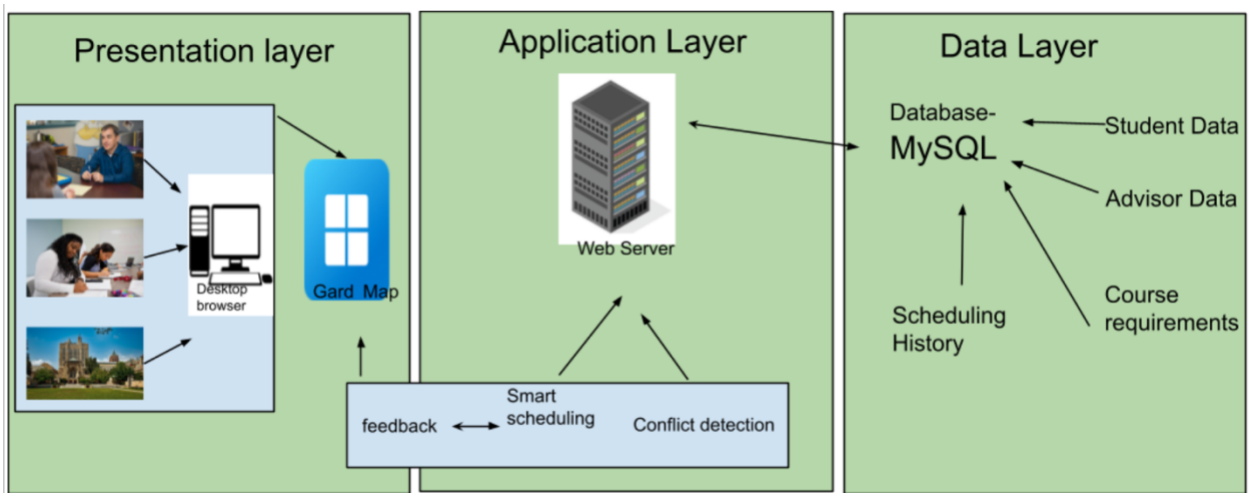
The GradMap prototype shall be developed locally and remotely as a web application with a database hosted by MySQL and accessible using MySQL Workbench. The prototype will run solely off mock data due to the ability to retrieve real student data from a university. The mock data will still be able to run the prototype effectively for demonstrative purposes. This is the only difference from a real-world application. The application will have two interfaces for students and advisors.

2.1 Product Perspective

GradMap is a web-based application composed of three primary system layers: the frontend interface, the backend application server, and a relational database. These layers work together to provide the functionality of the system and support communication between users and stored academic data. The system includes separate interfaces for students, advisors, and administrators to accommodate their different roles and responsibilities. Students interact with the system through a web interface where they can generate academic schedules and track their progress toward completing their degree requirements. Advisors use the system to review student-generated schedules and provide feedback or approval to help ensure students remain on track. Administrators manage the system's core data, including course information, degree requirements, and user accounts.

Figure 1

GradMap Prototype Major Functional Component Diagram



2.2 Product Functions

GradMap provides several core system functions that support students, advisors, and administrators. For students, the system includes secure user authentication, the ability to view degree requirements for their academic program, and tools for generating and editing course schedules. Students can plan courses by semester and track their progress toward completing their degree requirements. Advisors have access to features that allow them to securely log into the system, view the schedules of their assigned students, review academic plans, approve schedules or suggest modifications, and leave notes to guide students in adjusting their plans. Administrators are responsible for maintaining the system's academic data and have access to features that allow them to manage the course catalog and update degree requirement information within the system.

2.3 User Characteristics

The GradMap system supports three primary user groups: students, advisors, and administrators. Students are the primary users of the system and are expected to have basic computer literacy. They use GradMap to generate academic plans, organize courses by semester, and monitor their progress toward completing their degree requirements. Advisors are typically university faculty members or academic advisors who use the system to review and approve student schedules. These users are expected to have moderate technical experience and use the platform to provide guidance and feedback on student academic plans. Administrators are technical or administrative staff responsible for maintaining the system's core data and configuration. Their responsibilities include managing course information, updating degree

requirements, and maintaining user accounts, and they are expected to have a higher level of technical proficiency when interacting with the system.

2.4 Constraints

The GradMap system must operate within several technical and environmental constraints. The system must function within a standard web browser environment so that users can access it without requiring specialized software. Backend services must integrate with a relational database system to store and manage user data, course information, and degree requirements. Additionally, the system would have to comply with university data security and privacy guidelines to ensure that student and academic information is handled securely. Development of the system must also use technologies that are approved for the course project environment.

2.5 Assumptions and Dependencies

The GradMap system operates under several assumptions and dependencies. It is assumed that students and advisors will have reliable internet access to use the web-based application. It is also assumed that the course catalog data stored in the system is accurate and regularly maintained by administrators. Additionally, all users are expected to authenticate using secure login credentials to access their respective system features. The system depends on several technologies to function properly, including a backend server framework such as Node.js or an equivalent platform to handle application logic, and a relational database system such as MySQL to store and manage system data. The frontend must be compatible with standard web

browsers to allow users to access the system interface. Finally, secure password hashing libraries are required to protect user authentication data and maintain system security.